

Apple

Apple has been designing their own SoCs to power their own devices for quite some time now with the latest iteration A10 being one of the most powerful chips on the market currently. Designed in-house specifically for iPhones, these chips are manufactured either by Samsung or TSMC (Taiwan Semiconductor Manufacturing Company Limited).

A10 Fusion

A10 Fusion is the SoC that powers the iPhone 7 and 7 Plus and is the first quad core chip produced by Apple. The CPU features 2 high powered cores which are used for demanding applications and tasks like gaming and video editing, while the remaining two less powerful and more energy efficient cores are reserved for other menial tasks like email, texting and editing/ creating documents. These cores are setup in a configuration similar to the ARM big.LITTLE architecture.



Apple A10 Fusion is one of the fastest SoCs

Though a major difference between the A10 Fusion and other SoCs based on big.Little architecture like the Snapdragon 820, is that only one pair of cores (either the high powered or the low powered) can be active at one time making the A10 a dual core chip effectively.

According to Apple's claims, their latest chip is 40% faster and consumes 20% less energy than its predecessor, the Apple A9.

The GPU featured on the A10 is a 6 core chip that is 50% faster than its predecessor and consumes only 70% of the power.

Primary CPU	4x Apple Twister cores
CPU architecture	ARM big.LITTLE
CPU structure	RISC
DSP type	-
GPU	Improved PowerVR GT7600
Year Released	2016
Devices	iphone 7, iphone 7 plus